QUESTION ANALYSIS

EEE475

RF and Microwave Engineering

Question Analysis

Sessions 2020–21 to 2018–19 *3 papers, 77 questions, 138 formulas*

INSIDE

• At a glance • Heat map • Priority list • 12 topic pages • Derivations • Repeats • Styles • Study order

Prepared by Nur Mohammad

October 2026

Contents

1 At a glance **3**

2 Heat map **3**

3 Priority list **4**

4 Topic pages **5**

4.1 EM waves . . . 5
4.2 Transmission lines . . . 5
4.3 Smith chart . . . 8
4.4 Waveguides and microstrip . . . 9
4.5 Network analysis . . . 10
4.6 Impedance matching . . . 11
4.7 Resonators . . . 12
4.8 RF components and filters . . . 13
4.9 Amplifier design . . . 14
4.10 Microwave tubes and devices . . . 15
4.11 Microwave systems . . . 15
4.12 Antennas . . . 18

5 Derivations **22**

6 Repeat tracker **23**

7 Question style **23**

8 Study order **24**

Free shared copy · Nur Mohammad

1. At a glance:

This analysis covers **3 papers** (2020-21, 2019-20, 2018-19), with 77 questions and 452 marks offered. It shows what past papers asked. It does not predict the next paper, and with so few papers a pattern can easily be chance.

A short-notes question is split across its notes, each taking an equal share of the marks, so some marks are not whole numbers.

The five topics with the most marks:

Topic	Marks	Share	Questions	Top subtopics (marks)
Antennas	145	32 %	23	Hertzian dipole (25); Linear arrays (16); Radiation from accelerating charges (13)
Transmission lines	86	19 %	13	Terminated lines (30); Line equations (17); Line constants (11)
Impedance matching	43	10 %	7	Single-stub matching (28); Quarter-wave transformer (15)
Microwave systems	42	9 %	8	Radar (13); Radio link budget (11); Receivers and mixers (6)
Smith chart	28	6 %	4	Constant-r circles (18); Circle properties (6); Reading impedance (4)

In the topic pages, a formula name in bold is a key formula, the number after a subtopic is its marks offered, and a blue symbol in an equation links to its meaning in the symbol list, where the arrow leads back to the equation.

2. Heat map:

Marks offered, by topic and paper, in book order. A darker cell means more marks. Click a topic to open its page.

Topic and subtopic	2020-21	2019-20	2018-19	All
EM waves	**6**	·	·	**6**
Plane waves	**6**	·	·	6
Transmission lines	**31**	**34**	**21**	**86**
Terminated lines	**12**	**4**	**14**	30
Line equations	**8**	**9**	·	17
Line constants	**11**	·	·	11
Lossy lines	·	**10**	·	10
Open and short tests	·	·	**7**	7
Slotted-line measurement	·	**7**	·	7
Reflection coefficient	·	**2**	·	2
Standing wave ratio	·	**2**	·	2
Smith chart	**12**	**12**	**4**	**28**
Constant-r circles	**12**	**6**	·	18
Circle properties	·	**6**	·	6
Reading impedance	·	·	**4**	4
Waveguides and microstrip	**12**	**9**	**6**	**27**
Microstrip	**5**	·	**6**	11
Rectangular waveguide modes	**7**	·	·	7
TM waves	·	**6**	·	6
Circular waveguide modes	·	**3**	·	3
Network analysis	·	**11**	·	**11**
S-parameters	·	**6**	·	6
Signal flow graphs	·	**5**	·	5
Impedance matching	**19**	**10**	**14**	**43**

Topic and subtopic	2020-21	2019-20	2018-19	All
Single-stub matching	**14**	**6**	**8**	28
Quarter-wave transformer	**5**	**4**	**6**	15
Resonators	**3**	**2**	·	**5**
Cavity resonators	**3**	**2**	·	5
RF components and filters	·	**12**	**12**	**24**
RF behaviour of L and C	·	**6**	**6**	12
Two-port parameters	·	·	**6**	6
Filters	·	**6**	·	6
Amplifier design	·	**14**	**12**	**26**
Constant-gain circles	·	**9**	·	9
Oscillators	·	·	**7**	7
Stability	·	·	**5**	5
Constant-gain condition	·	**5**	·	5
Microwave tubes and devices	**9**	·	·	**9**
Klystron	**2.25**	·	·	2.25
Magnetron	**2.25**	·	·	2.25
Gunn diode	**2.25**	·	·	2.25
Travelling wave tube	**2.25**	·	·	2.25
Microwave systems	**16**	**9**	**17**	**42**
Radar	·	**7**	**6**	13
Radio link budget	·	·	**11**	11
Receivers and mixers	**6**	·	·	6
AM power	**5**	·	·	5
EIRP	**5**	·	·	5
RF hazards	·	**2**	·	2
Antennas	**42**	**39**	**64**	**145**
Hertzian dipole	**25**	·	·	25
Linear arrays	·	**7**	**9**	16
Radiation from accelerating charges	**13**	·	·	13
Antenna aperture	·	**6**	**6**	12
Near and far field	·	**7**	**5**	12
Pattern calculations	·	**5**	**6**	11
Remote sensing	·	**5**	**5**	10
Directivity and beam area	·	**9**	·	9
Magnetic dipole	·	·	**8**	8
Horn antennas	·	·	**7**	7
Paraboloid reflectors	·	·	**6**	6
Effective height and area	**4**	·	·	4
Reciprocity	·	·	**3**	3
Image theory	·	·	**3**	3
Polarization	·	·	**3**	3
Yagi-Uda antenna	·	·	**3**	3

3. Priority list:

Must know: Antennas (145), Transmission lines (86), Impedance matching (43), Microwave systems (42), Smith chart (28), Waveguides and microstrip (27)

Likely: Amplifier design (26), RF components and filters (24), Resonators (5)

Less often: Network analysis (11), Microwave tubes and devices (9), EM waves (6)

The rule: Must know means asked in every paper. Likely means asked in at least half the papers. Less often means asked in fewer than half. Numbers in brackets are marks offered, and each list runs from most marks to fewest.

4. Topic pages:

4.1. EM waves:

6 marks, 1 question, in 1 of 3 papers. Book: Pozar Ch. 1; Sadiku Ch. 10.

Most often numerical (1 of 1). Subtopics: Plane waves.

Formulas:

Subtopic	Formula	Equation	Used in
Plane waves (6)	**Wave speed**	$u = \dfrac{c}{\sqrt{\mu_r \epsilon_r}}$	20-21 3(a)
	Intrinsic impedance	$\eta = \eta_0 \sqrt{\dfrac{\mu_r}{\epsilon_r}}$	20-21 3(a)
	Wavelength	$\lambda = \dfrac{2\pi}{\beta}$	20-21 3(a)

Symbols:

μ_r, ϵ_r = relative permeability and permittivity ↩ η = ratio of E to H (Ω) ↩ λ = wavelength (m) ↩

Questions:

No.	Paper	Marks	Type	Question
Plane waves				
1	20-21 3(a)	6	numerical	A plane wave in a nonmagnetic medium has $\vec{E} = 50\sin(10^8 t + 2z)\hat{a}_y$ V/m. Find: (i) The direction of wave propagation (ii) λ, f, and ϵ_r (iii) $\vec{H}$

4.2. Transmission lines:

86 marks, 13 questions, in 3 of 3 papers. Book: Pozar Ch. 2; Sadiku Ch. 11.

Most often numerical (6 of 13). Subtopics: Terminated lines; Open and short tests; Lossy lines; Reflection coefficient; Standing wave ratio; Line equations; Slotted-line measurement; Line constants.

Formulas:

Subtopic	Formula	Equation	Used in
Terminated lines (30)	**Reflection coefficient**	$\Gamma_L = \dfrac{Z_L - Z_0}{Z_L + Z_0}$	20-21 1(c), 19-20 6(b), 18-19 3(c)
	Standing wave ratio	$S = \dfrac{1 + \lvert\Gamma\rvert}{1 - \lvert\Gamma\rvert}$	20-21 1(c), 19-20 4(a), 18-19 3(c)
	Voltage maximum	$d_{max} = \dfrac{\theta\lambda}{4\pi}$	18-19 3(c)
	Input impedance	$Z_{in} = Z_0 \dfrac{Z_L + jZ_0 \tan\beta l}{Z_0 + jZ_L \tan\beta l}$	18-19 3(c)

Free shared copy · Nur Mohammad

Subtopic	Formula	Equation	Used in
	Short-circuited line	$V = -2jV_0^+ \sin\beta z,\ I = \dfrac{2V_0^+}{Z_0}\cos\beta z$	18-19 4(c)
	Short and open input	$Z_{sc} = jZ_0\tan\beta l,\ Z_{oc} = -jZ_0\cot\beta l$	19-20 6(c), 18-19 4(c), 18-19 5(d)
	Load from Γ	$Z_L = Z_0\dfrac{1+\Gamma}{1-\Gamma}$	20-21 1(c)
	Input impedance	$Z_{in}(l) = \left(\dfrac{Z_L - jZ_0\tan\beta l}{Z_0 - jZ_L\tan\beta l}\right)Z_0$	20-21 1(c)
Open and short tests (7)	Z_0 from two measurements	$Z_0 = \sqrt{Z_{oc}Z_{sc}}$	19-20 6(c), 18-19 5(d)
	βl from two measurements	$\tan^2\beta l = -Z_{sc}/Z_{oc}$	18-19 5(d)
Lossy lines (10)	Lossy input impedance	$Z_{in} = Z_0\dfrac{Z_L + Z_0\tanh\gamma l}{Z_0 + Z_L\tanh\gamma l}$	19-20 2(d)
	γ from a short	$\gamma l = \dfrac{1}{2}\ln\dfrac{1+T}{1-T},\ T = \tanh\gamma l = \dfrac{Z_{sc}}{Z_0}$	19-20 2(d)
Reflection coefficient (2)	Along the line	$\Gamma(z) = \Gamma_L e^{2\gamma z},\ \Gamma(d) = \Gamma_L e^{-j2\beta d}$	19-20 4(a)
	Return loss	$RL = -20\log_{10}\lvert\Gamma\rvert$ dB	19-20 4(a)
Line equations (17)	Telegrapher's equations	$\dfrac{dV}{dz} = -(R+j\omega L)\,I,\ \dfrac{dI}{dz} = -(G+j\omega C)\,V$	20-21 1(a), 19-20 6(b)
	General solution	$V(z) = V_0^+e^{-\gamma z} + V_0^-e^{\gamma z}$	20-21 1(a), 19-20 6(b)
	Line current	$I(z) = \dfrac{1}{Z_0}\left(V_0^+e^{-\gamma z} - V_0^-e^{\gamma z}\right)$	19-20 6(b)
	Propagation constant	$\gamma = \alpha + j\beta = \sqrt{(R+j\omega L)(G+j\omega C)}$	20-21 1(a), 19-20 6(b)
	Characteristic impedance	$Z_0 = \dfrac{R+j\omega L}{\gamma} = \sqrt{\dfrac{R+j\omega L}{G+j\omega C}}$	20-21 3(b), 19-20 6(b)
	Low-loss line	$\alpha \approx \dfrac{R}{2Z_0} + \dfrac{GZ_0}{2},\ \beta \approx \omega\sqrt{LC},\ Z_0 \approx \sqrt{\dfrac{L}{C}}$	20-21 1(a)
Slotted-line measurement (7)	Phase from the minimum	$\theta_\Gamma = 2\beta d_{min} - \pi$	19-20 6(d)
	Impedance at extrema	$Z_{min} = \dfrac{Z_0}{S},\ Z_{max} = Z_0S$	19-20 6(d)
	Load from SWR	$Z_L = Z_0\dfrac{1 - jS\tan\beta d_{min}}{S - j\tan\beta d_{min}}$	19-20 6(d)
Line constants (11)	Conductor loss	$\alpha_c = \dfrac{R}{2Z_0},\ \alpha_d = \alpha - \alpha_c$	20-21 1(b)
	Shunt conductance	$G = \dfrac{2(\alpha - \alpha_c)}{Z_0}$	20-21 1(b)
	Phase velocity	$u_p = \dfrac{\omega}{\beta}$	20-21 3(b)

Free shared copy · Nur Mohammad

Symbols:

Γ_L = reflection coefficient at the load ↩

Z_L = load impedance (Ω) ↩

Z_0 = line impedance (Ω); characteristic impedance (Ω) ↩

Γ = reflection coefficient at the load ↩

S = SWR, $1 \leq S < \infty$ ↩

θ = angle of Γ_L (rad) ↩

d_{max} = distance from the load ↩

l = line length (m); distance from the load (m) ↩

β = phase constant (rad/m) ↩

V_0^+ = incident amplitude (V)

Z_{oc}, Z_{sc} = measured input impedances (Ω) ↩

d = distance from the load (m), lossless ↩

$|\Gamma|^2$ = fraction of power reflected

R, L, G, C = line constants per metre (Ω/m, H/m, S/m, F/m)

R = series resistance (Ω/m) ↩

L = series inductance (H/m) ↩

G = shunt conductance (S/m) ↩

C = shunt capacitance (F/m) ↩

V_0^+, V_0^- = incident and reflected amplitudes (V); forward and backward amplitudes (V) ↩

α = attenuation constant (Np/m) ↩

d_{min} = distance of the first voltage minimum (m) ↩

α_c, α_d = conductor and dielectric parts of α (Np/m) ↩

u_p = phase velocity (m/s) ↩

Questions:

No.	Paper	Marks	Type	Question
Terminated lines				
1	20-21 1(c)	12	derivation	In Fig. 1, a lossless transmission line is shown where, $V(z) = V_0^+(e^{-j\beta z} + \Gamma e^{j\beta z})$, $I(z) = \frac{V_0^+}{Z_0}(e^{-j\beta z} - \Gamma e^{j\beta z})$, Γ is the reflection coefficient at point $z = 0$, Z_0 is the characteristics impdedance of the line, Z_L is the load impedance, and β is the phase constant. Show that (i) $\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$ (ii) $Z_{in}(l) = \left(\frac{Z_L - jZ_0 \tan \beta l}{Z_0 - jZ_L \tan \beta l}\right) Z_0$
2	19-20 6(c)	4	theory	Briefly describe input impedance for open circuit termination and short circuit termination.
3	18-19 3(c)	7	numerical	A lossless transmission line length 0.434λ and characteristic impedance 100 Ω is terminated in an impedance $260 + j180$ Ω. Find the voltage reflection coefficient, standing wave ratio and the location of maximum voltage on the line.
4	18-19 4(c)	7	derivation	Derive the equation for a short circuit terminated transmission line and sketch voltage, current and impedance as a function of line length.
Open and short tests				
5	18-19 5(d)	7	numerical	A lossless transmission line of length 1.5 m, which is less than quarter wavelength has open circuit impedance $-j54.6$ Ω and short circuit impedance $j103$ Ω. Find Z_o and γ. Also find the input impedance if the line length is doubled and the length of line if it is to appear as an open circuit at the input.
Lossy lines				
6	19-20 2(d)	10	numerical	The input impedance of a short-circuited lossy transmission line of length 2 m and characteristics impedance 75 Ω is $15 + j120$ Ω. Find α and β of the line. Determine the input impedance if the sort-circuit *[sic: should be short-circuit]* is replaced by a load impedance $Z_L = 37 + j45$ Ω.
Reflection coefficient				
7	19-20 4(a)i	2	theory	Write short notes on the following topic (i) Reflection coefficient
Standing wave ratio				
8	19-20 4(a)ii	2	theory	Write short notes on the following topic (ii) Standing wave ratio
Line equations				
9	20-21 1(a)	8	derivation	From the lumped-element equivalent circuit, derive the expression of the propagation constant in terms of series resistance R, series inductance L, shunt conductance G, and shunt capacitance C per unit length. Deduce this expression to low-loss transmission line.
10	19-20 6(b)	9	derivation	From the general equation of transmission line find the expressions for characteristic impedance and reflection coefficient.

Free shared copy · Nur Mohammad

No.	Paper	Marks	Type	Question
Slotted-line measurement				
11	19-20 6(d)	7	numerical	The standing wave ratio on a lossless 50 Ω transmission line terminated in an unknown load impedance is found to be 2. The distance between the successive voltage minima is 15 cm, and the first minimum is located at 5 cm from the load. Determine reflection coefficient, load impedance and equivalent length and terminating resistance of a line such that the input impedance is equal to Z_L.
Line constants				
12	20-21 1(b)	5	numerical	If the propagation constant of a transmission line is $50\angle 89.38°$ m^{-1}, $R = 4$ Ω m^{-1}, $L = 0.5$ μH m^{-1}, $C = 200$ pF m^{-1}, calculate the value of shunt conductance with proper unit.
13	20-21 3(b)	6	numerical	A telephone line has $R = 30$ Ω km^{-1}, $L = 100$ mH km^{-1}, $G = 0$, and $C = 20$ μF km^{-1}. At $f = 1$ kHz, obtain: (i) The characteristic impedance of the line (ii) The propagation constant (iii) The phase velocity

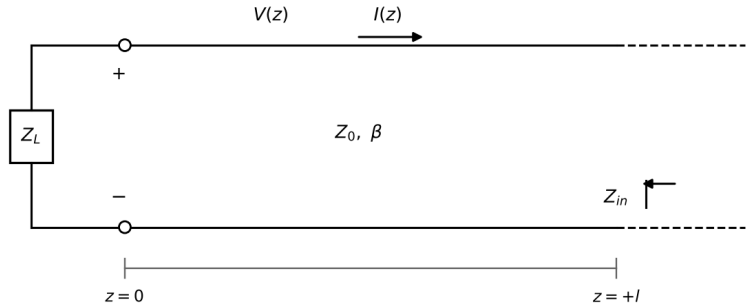


Figure 1: 20-21 1(c): Fig. 1 of the paper (redrawn)

4.3. Smith chart:

28 marks, 4 questions, in 3 of 3 papers. Book: Ludwig and Bretchko Ch. 3.

Most often theory (2 of 4). Subtopics: Reading impedance; Constant-r circles; Circle properties.

Formulas:

Subtopic	Formula	Equation	Used in
Reading impedance (4)	**Input impedance**	$Z_{in} = Z_0 \dfrac{Z_L + jZ_0 \tan \beta l}{Z_0 + jZ_L \tan \beta l}$	18-19 3(b)
Constant-r circles (18)	Impedance and Γ	$z = r + jx = \dfrac{1+\Gamma}{1-\Gamma}$	19-20 2(a)
	Constant-r circle	$\left(u - \dfrac{r}{1+r}\right)^2 + v^2 = \left(\dfrac{1}{1+r}\right)^2$	20-21 2(a), 19-20 2(a)
	Normalized impedance	$z = \dfrac{Z}{Z_0} = r + jx,\ \Gamma = \dfrac{z-1}{z+1}$	20-21 2(a)
Circle properties (6)	**Constant-x circle**	$(u-1)^2 + \left(v - \dfrac{1}{x}\right)^2 = \left(\dfrac{1}{x}\right)^2$	20-21 2(a), 19-20 2(b)

Free shared copy · Nur Mohammad

Symbols:

l = line length (m) ↩

β = phase constant (rad/m) ↩

r, x = normalized resistance and reactance ↩

u, v = real and imaginary parts of Γ ↩

Questions:

No.	Paper	Marks	Type	Question
Reading impedance				
1	18-19 3(b)	4	theory	Write the steps to determine line input impedance using Smith Chart.
Constant-r circles				
2	20-21 2(a)	12	derivation repeats 19-20 2(a)	Derive the equation of the constant resistance and constant reactance circle of a Smith Chart. You should start form the relation between the reflection coefficient and the normalized load.
3	19-20 2(a)	6	derivation	Show that the normalized resistances represent circles in reflection coefficient plane.
Circle properties				
4	19-20 2(b)	6	theory	Write the properties of r-circles and x-circles of a Smith chart.

4.4. Waveguides and microstrip:

27 marks, 5 questions, in 3 of 3 papers. Book: Pozar Ch. 3; Sadiku Ch. 12.

Most often theory (3 of 5). Subtopics: Microstrip; Circular waveguide modes; TM waves; Rectangular waveguide modes.

Formulas:

Subtopic	Formula	Equation	Used in
Circular waveguide modes (3)	TE wave impedance	$Z_{TE} = \dfrac{k\eta}{\beta}$	19-20 2(c)
	Circular guide cutoff	$f_c = \dfrac{p\,c}{2\pi a\sqrt{\epsilon_r}}$	19-20 2(c)
TM waves (6)	TM field	$E_z = E_0 \sin\dfrac{m\pi x}{a} \sin\dfrac{n\pi y}{b}\, e^{-j\beta z}$	19-20 5(a)
	Cutoff wave number	$k_c = \sqrt{\left(\dfrac{m\pi}{a}\right)^2 + \left(\dfrac{n\pi}{b}\right)^2},\ k_c^2 = k^2 - \beta^2$	19-20 5(a)
	Phase constant	$\beta = \sqrt{k^2 - k_c^2} = k\sqrt{1 - \left(\dfrac{f_c}{f}\right)^2}$	19-20 5(a)
	Cutoff frequency	$f_{c,mn} = \dfrac{1}{2\sqrt{\mu\epsilon}}\sqrt{\left(\dfrac{m}{a}\right)^2 + \left(\dfrac{n}{b}\right)^2}$	20-21 3(d), 19-20 5(a)
	TM wave impedance	$Z_{TM} = \dfrac{E_x}{H_y} = \dfrac{\beta}{\omega\epsilon} = \eta\sqrt{1 - \left(\dfrac{f_c}{f}\right)^2}$	19-20 5(a)

Symbols:

a = radius (m)

a, b = guide width and height (m) ↩

$k = \omega\sqrt{\mu\epsilon}$ = wave number (rad/m)

m, n = mode indices ↩

Questions:

No.	Paper	Marks	Type	Question
Microstrip				
1	20-21 6(a)	5	theory	What are the uses of a microstrip line? Mention its advantages. Describe the type of losses in a microstrip line.
2	18-19 4(b)	6	theory repeats 20-21 6(a)	What is microstrip antenna? Mention some of the applications of microstrip line.
Circular waveguide modes				
3	19-20 2(c)	3	theory	Describe how TE mode is different from TM mode in a circular waveguide.
TM waves				
4	19-20 5(a)	6	derivation	Derive the equations for wave impedance, cut-off frequency, and phase constant for a transverse magnetic wave.
Rectangular waveguide modes				
5	20-21 3(d)	7	numerical	A rectangular waveguide with dimensions $a = 2.5$ cm, $b = 1$ cm is to operate below 15.1 GHz. How many TE and TM modes can the waveguide transmit if the guide is filled with a medium characterized by $\sigma = 0$, $\epsilon = 4\epsilon_0$, $\mu_r = 1$? Calculate the cutoff frequencies of the modes.

4.5. Network analysis:

11 marks, 2 questions, in 1 of 3 papers. Book: Pozar Ch. 4.

Most often theory (1 of 2). Subtopics: S-parameters; Signal flow graphs.

Formulas:

Subtopic	Formula	Equation	Used in
S-parameters (6)	Power waves	$a_n = \dfrac{V_n^+}{\sqrt{Z_0}},\ b_n = \dfrac{V_n^-}{\sqrt{Z_0}}$	19-20 3(a)
	S-parameters	$[b] = [S][a],\ S_{ij} = \left.\dfrac{b_i}{a_j}\right\|_{a_k=0,\, k\neq j}$	19-20 3(a)
	Ideal circulator	$S_{21} = S_{32} = S_{13} = 1$, all other $S_{ij} = 0$	19-20 3(a)
Signal flow graphs (5)	**Input impedance**	$Z_{in} = Z_0 \dfrac{Z_L + jZ_0 \tan\beta l}{Z_0 + jZ_L \tan\beta l}$	19-20 3(c)
	Input reflection	$\Gamma_{in} = \Gamma_L e^{-j2\beta l}$	19-20 3(c)
	Flow graph with the source	$a_1 = \dfrac{b_s}{1 - \Gamma_S \Gamma_L e^{-j2\beta l}}$	19-20 3(c)

Symbols:

a_n, b_n = incident and outgoing waves at port n ($\sqrt{\text{W}}$) ↩

l = line length (m), lossless ↩

b_s = source wave ↩

Γ_S = source reflection coefficient ↩

Questions:

No.	Paper	Marks	Type	Question
S-parameters				
1	19-20 3(a)	6	theory	Why we use S-parameters for microwave analysis? Define S-matrix for 3 port network with appropriate example.
Signal flow graphs				
2	19-20 3(c)	5	derivation	For the figure shown in Fig. 1 draw the flow graph and derive the input impedance formula at port 1 from the signal flow-graph representation.

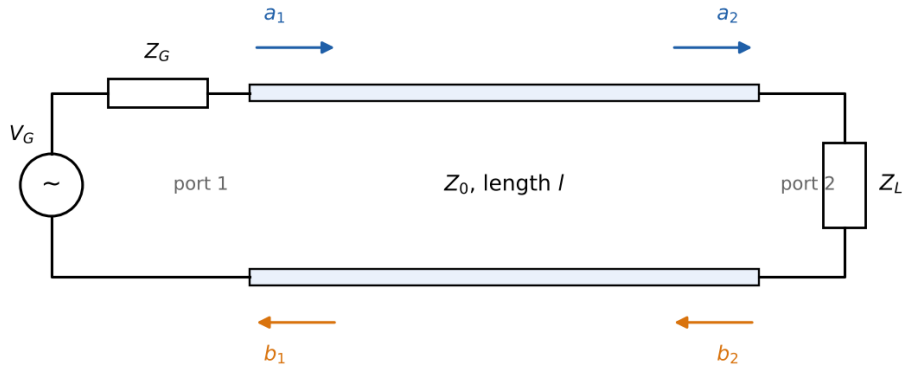


Figure 2: 19-20 3(c): Figure 1 of the paper (redrawn)

4.6. Impedance matching:

43 marks, 7 questions, in 3 of 3 papers. Book: Pozar Ch. 5.

Most often design (4 of 7). Subtopics: Single-stub matching; Quarter-wave transformer.

Formulas:

Subtopic	Formula	Equation	Used in
Single-stub matching (28)	Stub position	$t = \tan\beta d = \dfrac{X_L \pm \sqrt{R_L[(Z_0 - R_L)^2 + X_L^2]/Z_0}}{R_L - Z_0}$	19-20 4(d), 18-19 3(d)
	Short-circuited stub	$\dfrac{l}{\lambda} = \dfrac{1}{2\pi}\tan^{-1}\dfrac{Y_0}{B}$	20-21 2(b), 20-21 3(c), 19-20 4(d), 18-19 3(d)
	Susceptance to cancel	$B = \dfrac{R_L^2 t - (Z_0 - X_L t)(X_L + Z_0 t)}{Z_0\,[R_L^2 + (X_L + Z_0 t)^2]}$	19-20 4(d)
Quarter-wave transformer (15)	**Quarter-wave transformer**	$Z_1 = \sqrt{Z_0 R_L}$	20-21 2(c), 19-20 4(b), 18-19 4(d)

Symbols:

d = distance from the load (m) ↩

R_L, X_L = load resistance and reactance (Ω) ↩

B = susceptance to cancel (S); line susceptance at the stub (S) ↩

l_s = stub length (m)

Z_1 = impedance of the $\lambda/4$ section (Ω) ↩

R_L = real load (Ω); resistive load (Ω) ↩

Questions:

No.	Paper	Marks	Type	Question
Single-stub matching				
1	20-21 2(b)	8	design repeats 19-20 4(d)	A 75 Ω lossless transmission line is connected to a load impedance of $(35 - j50)$ Ω. Find the length position, l_p, and the length, l_s of a short-circuited stub required to match the load. You need to show two possible solutions using Smith Chart.
2	20-21 3(c)	6	design	An antenna with an impednace of $40 + j30$ Ω is to be matched to a 100 Ω lossless line with a shorted stub. Determine: (i) The required stub admittance (ii) The distance between the stub and the antenna (iii) The stub length (iv) The standing wave ratio on each segment of the system
3	19-20 4(d)	6	design	A 50 Ω transmission line is connected to a load impedance of $35 - j47.5$ Ω. Find the position and length of a short-circuited stub required to match the line.
4	18-19 3(d)	8	design repeats 19-20 4(d)	A 50 Ω transmission line is connected to a load impedance $35 - j47.5$ Ω. Find the position and length of a short circuited stub required to match the line.
Quarter-wave transformer				
5	20-21 2(c)	5	theory repeats 19-20 4(b)	Explain how quarter-wave transformer can be used for impdeance matching. Mention its limitations.
6	19-20 4(b)	4	theory	Show how quarter wavelength line is used for matching.
7	18-19 4(d)	6	theory repeats 19-20 4(b)	How is quarter wavelength transmission line used for impedance matching?

4.7. Resonators:

5 marks, 2 questions, in 2 of 3 papers. Book: Pozar Ch. 6.

Most often theory (1 of 2). Subtopics: Cavity resonators.

Formulas:

Subtopic	Formula	Equation	Used in
Cavity resonators (5)	**Cavity resonance**	$f_{mnp} = \frac{c}{2\sqrt{\mu_r \epsilon_r}} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{d}\right)^2}$	20-21 6(c), 19-20 4(a)
	Quality factor	$Q = \omega \frac{\text{energy stored}}{\text{power lost}}$	19-20 4(a)

Symbols:

d = cavity length (m) ↩ p = third mode index ↩

Questions:

No.	Paper	Marks	Type	Question
Cavity resonators				
1	20-21 6(c)	3	numerical	A rectangular cavity resonator has dimensions of $a = 5$ cm, $b = 2$ cm, and $d = 15$ cm. Compute the resonant frequency of the dominant mode for a dielectric-field cavity of $\epsilon_r = 2.56$.
2	19-20 4(a)iv	2	theory	Write short notes on the following topic (iv) Microwave cavity resonators

Free shared copy · Nur Mohammad

4.8. RF components and filters:

24 marks, 4 questions, in 2 of 3 papers. Book: Ludwig and Bretchko Ch. 1, 4, 5.

Most often derivation (2 of 4). Subtopics: RF behaviour of L and C; Two-port parameters; Filters.

Formulas:

Subtopic	Formula	Equation	Used in
RF behaviour of L and C (12)	High-frequency resistor	$Z = j\omega(2L) + \dfrac{1}{1/R + j\omega C_a}$	18-19 4(a)
	Resistor corners	$f_c = \dfrac{1}{2\pi R C_a},\ f_0 = \dfrac{1}{2\pi\sqrt{2LC_a}}$	18-19 4(a)
	Ideal reactances	$X_L = 2\pi f L,\ X_C = \dfrac{1}{2\pi f C}$	19-20 6(a)
	Capacitor self-resonance	$f_r = \dfrac{1}{2\pi\sqrt{L_s C}}$	19-20 6(a)
	Inductor self-resonance	$f_r = \dfrac{1}{2\pi\sqrt{L C_s}}$	19-20 6(a)
Two-port parameters (6)	ABCD definition	$V_1 = AV_2 + BI_2,\ I_1 = CV_2 + DI_2$	18-19 5(a)
	High-pass ABCD	$A = 1 - \dfrac{1}{\omega^2 LC},\ B = \dfrac{1}{j\omega C},\ C = \dfrac{1}{j\omega L},\ D = 1$	18-19 5(a)
Filters (6)	**High-pass gain**	$H = \dfrac{V_2}{V_1} = \dfrac{j\omega RC}{1 + j\omega RC},\ f_c = \dfrac{1}{2\pi RC}$	19-20 5(b)
	Gain and phase	$\lvert H\rvert = \dfrac{f/f_c}{\sqrt{1 + (f/f_c)^2}},\ \angle H = 90° - \tan^{-1}\dfrac{f}{f_c}$	19-20 5(b)
	Series C in a Z_0 system	$S_{21} = \dfrac{j\omega(2Z_0C)}{1 + j\omega(2Z_0C)}$	19-20 5(b)

Symbols:

L = each lead's inductance (H) ↩ f = frequency (Hz) ↩ C_s = winding capacitance (F) ↩

C_a = end capacitance (F) ↩ L_s = lead inductance (H) ↩ V_1, V_2 = input and output voltages ↩

Questions:

No.	Paper	Marks	Type	Question
RF behaviour of L and C				
1	19-20 6(a)	6	theory	Explain how radio frequency affects the inductor and capacitor reactance.
2	18-19 4(a)	6	diagram	Sketch the equivalent circuit of a high frequency resistor. Also plot a typical frequency versus impedance cure *[sic: should be curve]* of the resistor.
Two-port parameters				
3	18-19 5(a)	6	derivation	For high pass filter determine ABCD parameters of a two port system.
Filters				
4	19-20 5(b)	6	derivation	Find the forward voltage gain of a high-pass filter.

Free shared copy · Nur Mohammad

4.9. Amplifier design:

26 marks, 4 questions, in 2 of 3 papers. Book: Ludwig and Bretchko Ch. 9, 10; Pozar Ch. 12, 13.

Most often theory (3 of 4). Subtopics: Stability; Oscillators; Constant-gain condition; Constant-gain circles.

Formulas:

Subtopic	Formula	Equation	Used in
Stability (5)	Input reflection	$\Gamma_{in} = S_{11} + \dfrac{S_{12}S_{21}\Gamma_L}{1 - S_{22}\Gamma_L}$	18-19 5(b)
	Output stability circle	$C_L = \dfrac{(S_{22} - \Delta S_{11}^*)^*}{\lvert S_{22}\rvert^2 - \lvert\Delta\rvert^2},\ R_L = \left\lvert \dfrac{S_{12}S_{21}}{\lvert S_{22}\rvert^2 - \lvert\Delta\rvert^2} \right\rvert$	18-19 5(b)
	Input stability circle	$C_S = \dfrac{(S_{11} - \Delta S_{22}^*)^*}{\lvert S_{11}\rvert^2 - \lvert\Delta\rvert^2},\ R_S = \left\lvert \dfrac{S_{12}S_{21}}{\lvert S_{11}\rvert^2 - \lvert\Delta\rvert^2} \right\rvert$	18-19 5(b)
Oscillators (7)	Negative-resistance oscillator	$R_{in} + R_L = 0,\ X_{in} + X_L = 0$	18-19 5(c)
	Barkhausen condition	$\lvert A\beta\rvert = 1,\ \angle A\beta = 0°$	18-19 5(c)
Constant-gain condition (5)	Unilateral gain	$G_{TU} = G_S\, G_0\, G_L,\ G_0 = \lvert S_{21}\rvert^2$	19-20 3(b)
	Source gain	$G_S = \dfrac{1 - \lvert\Gamma_S\rvert^2}{\lvert 1 - S_{11}\Gamma_S\rvert^2}$	19-20 3(b)
	Normalized gain	$g_s = \dfrac{G_S}{G_{Smax}} = G_S(1 - \lvert S_{11}\rvert^2)$	19-20 3(b)
	Gain-circle centre	$d_{gs} = \dfrac{g_s S_{11}^*}{1 - \lvert S_{11}\rvert^2(1 - g_s)}$	19-20 3(b)
	Gain-circle radius	$r_{gs} = \dfrac{\sqrt{1 - g_s}\,(1 - \lvert S_{11}\rvert^2)}{1 - \lvert S_{11}\rvert^2(1 - g_s)}$	19-20 3(b)
Constant-gain circles (9)	**Maximum source gain**	$G_{Smax} = \dfrac{1}{1 - \lvert S_{11}\rvert^2}$	19-20 3(d)

Free shared copy · Nur Mohammad

No.	Paper	Marks	Type	Question

Symbols:

S_{ij} = S-parameters

β = feedback ↩

r_{gs} = radius ↩

A = gain ↩

d_{gs} = centre in the Γ_S plane ↩

Questions:

No.	Paper	Marks	Type	Question
Stability				
1	18-19 5(b)	5	theory	What are stability circles? Explain how it is used.
Oscillators				
2	18-19 5(c)	7	theory	Explain the working principle of negative resistance oscillator and feedback oscillator.
Constant-gain condition				
3	19-20 3(b)	5	theory	Explain the condition for constant gain for amplifier.
Constant-gain circles				
4	19-20 3(d)	9	design	A FET is operated at $f = 4$ GHz and is biased such that $s_{11} = 0.7\angle 125°$. It is assumed that the transistor is unconditionally stable. Find the maximum source gain G_{Smax} and sketch the constant gain circles for several values of G_S.

4.10. Microwave tubes and devices:

9 marks, 4 questions, in 1 of 3 papers. Book: Pozar Ch. 11.

Most often theory (4 of 4). Subtopics: Klystron; Magnetron; Gunn diode; Travelling wave tube.

Questions:

No.	Paper	Marks	Type	Question
Klystron				
1	20-21 5(d)i	2.25	theory	Write short notes (any three). (i) Klystron
Magnetron				
2	20-21 5(d)ii	2.25	theory	Write short notes (any three). (ii) Magnetron
Gunn diode				
3	20-21 5(d)iii	2.25	theory	Write short notes (any three). (iii) Gunn Diode Oscillator
Travelling wave tube				
4	20-21 5(d)iv	2.25	theory	Write short notes (any three). (iv) Travelling Wave Tube (TWT)

4.11. Microwave systems:

42 marks, 8 questions, in 3 of 3 papers. Book: Pozar Ch. 14.

Most often theory (3 of 8). Subtopics: Radar; Radio link budget; RF hazards; Receivers and mixers; AM power; EIRP.

Formulas:

Subtopic	Formula	Equation	Used in
Radar (13)	**Radar range**	$R_{max} = \left[\frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 P_{min}}\right]^{1/4}$	19-20 4(c), 18-19 2(b)
	Density at the target	$S_t = \frac{P_t G}{4\pi R^2}$	19-20 4(c)
	Radar equation	$P_r = \frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 R^4}$	19-20 4(c)

Free shared copy · Nur Mohammad

Subtopic	Formula	Equation	Used in
	Range from delay	$R = \frac{c\,t}{2}$	19-20 4(c)
Radio link budget (11)	**Directivity from aperture**	$D = \frac{4\pi A_e}{\lambda^2}$	18-19 6(c)
	Friis formula	$\frac{P_r}{P_t} = \frac{A_{et}A_{er}}{r^2\lambda^2} = G_tG_r\left(\frac{\lambda}{4\pi r}\right)^2$	18-19 2(c), 18-19 6(c)
	Noise power	$N = kT_{sys}B$	18-19 2(c)
	Link SNR	$\frac{S}{N} = \frac{P_tG_tG_r}{kT_{sys}B}\left(\frac{\lambda}{4\pi r}\right)^2$	18-19 2(c)
RF hazards (2)	Specific absorption rate	$\text{SAR} = \frac{\sigma_t\lvert E\rvert^2}{\rho}$	19-20 4(a)
	Safe distance	$r = \sqrt{\frac{\text{EIRP}}{4\pi S_{lim}}}$	19-20 4(a)
Receivers and mixers (6)	Mixer product	$\cos\omega_{RF}t\cos\omega_{LO}t = \frac{1}{2}\cos(\omega_{RF}-\omega_{LO})t + \frac{1}{2}\cos(\omega_{RF}+\omega_{LO})t$	20-21 5(a)
	Intermediate frequency	$f_{IF} = f_{RF} - f_{LO}$	20-21 5(a)
	Image frequency	$f_{image} = f_{LO} - f_{IF}$	20-21 5(a)
AM power (5)	AM signal	$s(t) = A_c[1+\mu\cos\omega_m t]\cos\omega_c t$	20-21 5(b)
	AM total power	$P_t = P_c\left(1+\frac{\mu^2}{2}\right)$	20-21 5(b)
EIRP (5)	Gain in dB	$G_{dB} = 10\log_{10}G,\ G = 10^{G_{dB}/10}$	20-21 5(c)
	EIRP	$\text{EIRP} = P_tG_t$	20-21 5(c)

Free shared copy · Nur Mohammad

No.	Paper	Marks	Type	Question

Symbols:

σ = radar cross-section (m^2) ↩

P_{min} = minimum detectable signal (W) ↩

P_t = transmitted power (W); transmitter power (W) ↩

G = antenna gain ↩

R = range (m) ↩

t = echo delay (s) ↩

A_e = effective aperture (m^2) ↩

A_{et}, A_{er} = apertures (m^2) ↩

k = Boltzmann's constant (J/K)

T_{sys} = system temperature (K) ↩

B = bandwidth (Hz) ↩

σ_t = tissue conductivity (S/m) ↩

ρ = tissue density (kg/m^3) ↩

S_{lim} = power-density limit (W/m^2) ↩

ω_{RF}, ω_{LO} = signal and local-oscillator frequencies (rad/s) ↩

A_c = carrier amplitude ↩

μ = modulation index ↩

P_c = carrier power (W) ↩

G_{dB} = gain in decibels (dBi) ↩

G_t = transmit antenna gain ↩

Questions:

No.	Paper	Marks	Type	Question
Radar				
1	19-20 4(c)	7	theory	Explain how the RADAR range equation is used to detect object.
2	18-19 2(b)	6	theory repeats 19-20 4(c)	Explain how the RADAR range equation is used to detect objects.
Radio link budget				
3	18-19 2(c)	5	derivation	Show the signal to noise ratio of a radio link is related to bandwidth and system temperature.
4	18-19 6(c)	6	derivation	Show that for antennas with same aperture the received to transmitted power ratio is inversely proportional to square of wavelength and distance between them.
RF hazards				
5	19-20 4(a)iii	2	theory	Write short notes on the following topic (iii) RF radiation hazards and safety standard
Receivers and mixers				
6	20-21 5(a)	6	numerical	An RF signal at $f = 100$ MHz is mixed with local oscillator signal at $f_{LO} = 90$ MHz. (i) Calculate the IF (intermediate frequency) output. (ii) Why are active filters used after mixing in a super-heterodyne receiver?
AM power				
7	20-21 5(b)	5	numerical	For a 1 kHz message signal modulating a 1 MHz carrier with 80% modulation index in AM, calculate the total transmitted power if carrier power is 100 W.
EIRP				
8	20-21 5(c)	5	numerical	For an antenna with gain 10 dB transmitting 10 W power at 2.4 GHz, calculate the effective isotropic radiated power (EIRP) of the antenna.

Free shared copy · Nur Mohammad

4.12. Antennas:

145 marks, 23 questions, in 3 of 3 papers. Book: Kraus; Sadiku Ch. 13.

Most often theory (12 of 23). Subtopics: Antenna aperture; Pattern calculations; Near and far field; Magnetic dipole; Remote sensing; Linear arrays; Reciprocity; Image theory; Polarization; Yagi-Uda antenna; Paraboloid reflectors; Horn antennas; Directivity and beam area; Hertzian dipole; Radiation from accelerating charges; Effective height and area.

Formulas:

Subtopic	Formula	Equation	Used in
Antenna aperture (12)	**Directivity from aperture**	$D = \dfrac{4\pi A_e}{\lambda^2}$	18-19 1(a), 18-19 6(d)
	Effective aperture	$A_e = \dfrac{P_r}{S},\ \varepsilon_{ap} = \dfrac{A_e}{A_p}$	19-20 1(b)
	Aperture and directivity	$D = \dfrac{4\pi A_e}{\lambda^2},\ A_e\,\Omega_A = \lambda^2$	19-20 1(b)
	Aperture beamwidth	$\text{HPBW} \approx \dfrac{51^\circ\,\lambda}{a}$	19-20 1(b)
Pattern calculations (11)	Power and beam area	$P_{rad} = \iint U\,d\Omega,\ \Omega_A = \dfrac{P_{rad}}{U_m}$	18-19 1(b)
	Directivity	$D = \dfrac{4\pi U_m}{P_{rad}} = \dfrac{4\pi}{\Omega_A}$	19-20 1(a), 18-19 1(b)
	Radiated power	$P_{rad} = \displaystyle\int_0^{2\pi}\int_0^{\pi} U \sin\theta\,d\theta\,d\phi$	19-20 1(d)
Near and far field (12)	Field regions	$r_1 = 0.62\sqrt{D^3/\lambda},\ r_2 = 2D^2/\lambda$	19-20 5(d), 18-19 1(c)
Magnetic dipole (8)	Magnetic dipole E	$E_\phi = \dfrac{\eta k^2 IS}{4\pi r}\sin\theta\left[1 + \dfrac{1}{jkr}\right]e^{-jkr}$	18-19 1(d)
	Magnetic dipole H_r	$H_r = \dfrac{jkIS}{2\pi r^2}\cos\theta\left[1 + \dfrac{1}{jkr}\right]e^{-jkr}$	18-19 1(d)
	Magnetic dipole H_θ	$H_\theta = -\dfrac{k^2 IS}{4\pi r}\sin\theta\left[1 + \dfrac{1}{jkr} - \dfrac{1}{(kr)^2}\right]e^{-jkr}$	18-19 1(d)
Remote sensing (10)	Brightness temperature	$T_B = e\,T_{phys}$	18-19 2(a)
	Received noise power	$P = kT_A B$	19-20 1(c), 18-19 2(a)
	Brightness temperature	$T_B = e\,T_{phys},\ e = 1 - \lvert\Gamma\rvert^2$	19-20 1(c)
	Antenna temperature	$T_A = \dfrac{1}{\Omega_A}\iint T_B\,P_n\,d\Omega$	19-20 1(c)
	Radiometer resolution	$\Delta T_{min} = \dfrac{T_{sys}}{\sqrt{B\tau}}$	19-20 1(c)
Linear arrays (16)	Phase difference	$\psi = \beta d\cos\theta + \delta$	19-20 5(c), 18-19 2(d)
	Array factor	$\lvert AF_n\rvert = \left\lvert\dfrac{\sin(N\psi/2)}{N\sin(\psi/2)}\right\rvert$	19-20 5(c), 18-19 2(d)

Free shared copy · Nur Mohammad

Subtopic	Formula	Equation	Used in
	Main beam	$\cos\theta_m = -\dfrac{\delta}{\beta d}$	19-20 5(c), 18-19 2(d)
	Nulls	$\theta_k = \cos^{-1}\left[\dfrac{1}{\beta d}\left(-\delta \pm \dfrac{2k\pi}{N}\right)\right]$	19-20 5(c), 18-19 2(d)
	Broadside beamwidths	$\text{FNBW} \approx \dfrac{2\lambda}{Nd},\ \text{HPBW} \approx 0.886\dfrac{\lambda}{Nd}$	18-19 2(d)
	Broadside beamwidths	$\text{FNBW} = 2\left[\dfrac{\pi}{2} - \cos^{-1}\dfrac{\lambda}{Nd}\right],\ \text{HPBW} \approx 0.886\,\dfrac{\lambda}{Nd}$	19-20 5(c)
	End-fire beamwidth	$\text{FNBW} = 2\cos^{-1}\left(1 - \dfrac{\lambda}{Nd}\right)$	19-20 5(c)
Polarization (3)	Polarization match	$F = \cos^2\gamma$	18-19 6(a)
Paraboloid reflectors (6)	Paraboloid shape	$r = \dfrac{2f}{1+\cos\theta}$	18-19 6(b)
	Paraboloid taper	$\dfrac{S(\theta)}{S(0)} = \left(\dfrac{1+\cos\theta}{2}\right)^2,\ \dfrac{E(\theta)}{E(0)} = \dfrac{1+\cos\theta}{2}$	18-19 6(b)
Horn antennas (7)	**Horn directivity**	$D \approx \dfrac{7.5A_p}{\lambda^2}$	18-19 6(d)
Directivity and beam area (9)	Directivity estimate	$D \approx \dfrac{41\,253}{\theta_{HP}\,\phi_{HP}}$	19-20 1(a)
	Gain	$G = k\,D$	19-20 1(a)
	Beam area	$\Omega_A = \iint P_n(\theta,\phi)\sin\theta\,d\theta\,d\phi \approx \theta_{HP}\,\phi_{HP}$	19-20 1(a)
	Effective height	$h_e = \dfrac{V_{oc}}{E} = \dfrac{1}{I_0}\displaystyle\int I(z)\,dz$	20-21 6(d), 19-20 1(a)
	Height and aperture	$A_e = \dfrac{h_e^2\,Z_0}{4R_r}$	19-20 1(a)
Hertzian dipole (25)	Vector potential	$A_r = \dfrac{\mu_0 Idl}{4\pi r}e^{-j\beta r}\cos\theta,\ A_\theta = -\dfrac{\mu_0 Idl}{4\pi r}e^{-j\beta r}\sin\theta$	20-21 4(a)
	Magnetic field	$\mathbf{H} = \dfrac{Idl}{4\pi}\sin\theta\left[\dfrac{j\beta}{r} + \dfrac{1}{r^2}\right]e^{-j\beta r}\,\hat{a}_\phi$	20-21 4(a)
	Field from **H**	$\mathbf{E} = \dfrac{1}{j\omega\epsilon}\nabla\times\mathbf{H},\ \dfrac{1}{j\omega\epsilon} = \dfrac{\eta}{j\beta}$	20-21 4(b)
	Radial E field	$E_r = \dfrac{\eta Idl}{2\pi}\cos\theta\left[\dfrac{1}{r^2} - \dfrac{j}{\beta r^3}\right]e^{-j\beta r}$	20-21 4(b)
	Polar E field	$E_\theta = \dfrac{\eta Idl}{4\pi}\sin\theta\left[\dfrac{j\beta}{r} + \dfrac{1}{r^2} - \dfrac{j}{\beta r^3}\right]e^{-j\beta r}$	20-21 4(b)
	Average Poynting vector	$\mathbf{P}_{avg} = \dfrac{1}{2}\text{Re}\,(\mathbf{E}\times\mathbf{H}^*)$	20-21 4(c)
	Far-field power density	$\mathbf{P}_{avg} = \dfrac{15\pi I_0^2}{r^2}\left(\dfrac{dl}{\lambda}\right)^2\sin^2\theta\,\hat{a}_r$	20-21 4(c)

Free shared copy · Nur Mohammad

Subtopic	Formula	Equation	Used in
	Radiated power	$P_{rad} = 40\pi^2 \left(\frac{dl}{\lambda}\right)^2 I_0^2$	20-21 4(c)
	Radiation resistance	$R_{rad} = \frac{2P_{rad}}{I_0^2} = 80\pi^2 \left(\frac{dl}{\lambda}\right)^2$	20-21 4(c)
Radiation from accelerating charges (13)	Retarded time	$t_r = t - \frac{R}{c}$	20-21 6(b)
	Retarded potentials	$V = \frac{1}{4\pi\epsilon_0}\int \frac{\rho(t_r)}{R}\,d\tau',\ \mathbf{A} = \frac{\mu_0}{4\pi}\int \frac{\mathbf{J}(t_r)}{R}\,d\tau'$	20-21 6(b)
	Jefimenko, E	$\mathbf{E} = \frac{1}{4\pi\epsilon_0}\int \left[\frac{\rho}{R^2}\hat{\mathbf{R}} + \frac{\dot{\rho}}{cR}\hat{\mathbf{R}} - \frac{\dot{\mathbf{J}}}{c^2 R}\right] d\tau'$	20-21 6(b)
	Jefimenko, B	$\mathbf{B} = \frac{\mu_0}{4\pi}\int \left[\frac{\mathbf{J}}{R^2} + \frac{\dot{\mathbf{J}}}{cR}\right] \times \hat{\mathbf{R}}\,d\tau'$	20-21 6(b)
Effective height and area (4)	**Effective area**	$A_e = \frac{P_{received}}{P_{inc}} = \frac{\lambda^2}{4\pi}G$	20-21 6(d)
	Area from height	$A_e = \frac{\eta\, h_e^2}{4R_{rad}}$	20-21 6(d)

Symbols:

A_e = effective aperture (m^2) ↩

P_r = received power (W) ↩

S = power density (W/m^2); loop area (m^2) ↩

A_p = physical aperture (m^2); mouth area (m^2) ↩

λ = wavelength (m) ↩

a = width of a uniform aperture (m) ↩

U = radiation intensity (W/sr) ↩

P_{rad} = total radiated power (W) ↩

$\sin\theta\, d\theta\, d\phi$ = element of solid angle (sr) ↩

D = largest dimension (m); largest antenna size (m) ↩

e = emissivity ↩

T_A = antenna temperature (K) ↩

k = Boltzmann's constant; radiation efficiency ↩

B = bandwidth (Hz) ↩

T_{phys} = physical temperature (K) ↩

T_{sys} = system noise temperature (K) ↩

τ = integration time (s) ↩

d = element spacing (m) ↩

δ = progressive feed phase (rad) ↩

N = number of elements ↩

2γ = angle on the Poincaré sphere

f = focal length (m) ↩

θ_{HP}, ϕ_{HP} = half-power beamwidths (deg) ↩

P_n = normalized power pattern ↩

V_{oc} = open-circuit voltage (V) ↩

I_0 = terminal current (A) ↩

E = incident field (V/m) ↩

R_r = radiation resistance (Ω) ↩

r, θ = spherical coordinates (m, rad) ↩

$\mathbf{H}$ = magnetic field (A/m) ↩

$\mathbf{P}_{avg}$ = power density (W/m^2) ↩

R_{rad} = radiation resistance (Ω) ↩

R = distance from source point to field point (m) ↩

ρ = charge density (C/m^3) ↩

$\mathbf{J}$ = current density (A/m^2) ↩

P_{inc} = incident power density (W/m^2) ↩

G = antenna gain ↩

Questions:

No.	Paper	Marks	Type	Question
Antenna aperture				
1	19-20 1(b)	6	theory	What is aperture of an Antenna? Show how antenna aperture can be set to increase directivity of an antenna.
2	18-19 1(a)	6	theory repeats 19-20 1(b)	What is aperture of an antenna? Show how antenna aperture can be set to increase directivity of an antenna.
Pattern calculations				
3	19-20 1(d)	5	numerical	Calculate the half power beamwidth, beamwidth between first nulls, radiated power and directivity of an antenna with power pattern $sin\theta sin\varphi$ for $0 \leq \theta \leq \pi,\ 0 \leq \varphi \leq \pi$.
4	18-19 1(b)	6	numerical repeats 19-20 1(d)	Calculate the half power beamwidth, beamwidth between first nulls, radiated power and directivity of an antenna with power pattern $\cos^2\theta \sin\varphi$ for $0 \leq \theta \leq \pi,\ 0 \leq \varphi \leq \pi$.
Near and far field				
5	19-20 5(d)	7	theory	Define near field and far field for an antenna.
6	18-19 1(c)	5	theory repeats 19-20 5(d)	Describe the field regions surrounding an antenna.
Magnetic dipole				
7	18-19 1(d)	8	derivation	For an elemental magnetic dipole derive the equation for electric field and magnetic field intensity.
Remote sensing				
8	19-20 1(c)	5	theory	Describe how passive remote sensing is used to detect object.
9	18-19 2(a)	5	theory repeats 19-20 1(c)	Describe how passive remote sensing is used to detect object.
Linear arrays				
10	19-20 5(c)	7	derivation	Derive the equation for a uniform linear array and write the conditions for main beam direction, width of main beam and null locations.
11	18-19 2(d)	9	derivation repeats 19-20 5(c)	Derive the equation for a uniform linear array and write the conditions for main beam direction, width of main beam and null locations.
Reciprocity				
12	18-19 3(a)i	3	theory	Write short notes on. (i) Lorentz reciprocity theorem.
Image theory				
13	18-19 3(a)ii	3	theory	Write short notes on. (ii) Image theory.
Polarization				
14	18-19 6(a)i	3	theory	Write short notes on the following topics. (i) Poincare Sphere
Yagi-Uda antenna				
15	18-19 6(a)ii	3	theory	Write short notes on the following topics. (ii) Yagi-Uda antenna.
Paraboloid reflectors				
16	18-19 6(b)	6	derivation	For a paraboloid-of-revolution with an isotropic point source what is the ratio of power density and field intensity ratio.
Horn antennas				
17	18-19 6(d)	7	derivation	For a pyramidal or rectangular horn show that directivity D is $D \approx 10\log\left(\frac{7.5A_p}{\lambda^2}\right)$.
Directivity and beam area				
18	19-20 1(a)	9	theory	Write short notes on the following topics (i) Directivity (ii) Effective Height (iii) Beam Area

Free shared copy · Nur Mohammad

No.	Paper	Marks	Type	Question
Hertzian dipole				
19	20-21 4(a)	8	derivation	The magnetic vector potential of a Hertzian dipole is given by: $\vec{A} = \frac{\mu_0 I dl}{4\pi r} e^{-j\beta r}\hat{z}$. Show that magnetic field, $H = \frac{Idl}{4\pi}\cos(\theta)\left[\frac{j\beta}{r} + \frac{1}{r^2}\right] e^{-j\beta r}\hat{\phi}$ *[sic: should be* $\sin(\theta)$*]*.
20	20-21 4(b)	12	derivation	Using the expression of H found in 4(a), show that the electric field component $E = \frac{\eta I dl}{4\pi}\cos(\theta)\left[\frac{j\beta}{r} + \frac{1}{r^2} - \frac{j}{\beta r^3}\right] e^{-j\beta r}\hat{\theta} + \frac{\eta I dl}{2\pi}\cos(\theta)\left[\frac{1}{r^2} - \frac{j}{\beta r^3}\right] e^{-j\beta r}\hat{r}$ *[sic: the* $\hat{\theta}$ *term should have* $\sin(\theta)$*]*
21	20-21 4(c)	5	derivation	Find the expression of average power density of Hertzian dipole in far field zone.
Radiation from accelerating charges				
22	20-21 6(b)	13	derivation	Derive Jefeminko's equation and from there, explain the necessity of accelerating charge for electromagnetic radiation.
Effective height and area				
23	20-21 6(d)	4	theory repeats 19-20 1(a)	What do you mean by effective height and effective area of an antenna?

5. Derivations:

Every derivation asked, in book order. Each outline gives the steps; the link opens the full working.

No.	Paper	Marks	Derivation	Outline
Transmission lines				
1	20-21 1(a)	8	Line equations	Apply KVL and KCL to a short section; let Δz go to zero; combine into the wave equation; read off γ; simplify for low loss.
2	20-21 1(c)	12	Terminated lines	Apply V and I at the load to find Γ; move a distance l along the line; form V/I; rewrite in tan βl.
3	19-20 6(b)	9	Line equations	Solve the line equations; take V and I as forward and backward waves; form V/I for Z_0; apply the load for Γ.
4	18-19 4(c)	7	Terminated lines	Apply $V = 0$ at $z = 0$, so $V_0^- = -V_0^+$; use Euler identities for V and I; divide for Z_{in}.
Smith chart				
5	20-21 2(a)	12	Constant-r circles	Write z = (1+Γ)/(1−Γ) with Γ = u+jv; split real and imaginary parts; complete the square for each circle.
6	19-20 2(a)	6	Constant-r circles	Write z = (1+Γ)/(1−Γ) with Γ = u+jv; take the real part; complete the square to get a circle.
Waveguides and microstrip				
7	19-20 5(a)	6	TM waves	Start from the wave equation for E_z; separate variables; apply the wall conditions; find k_c, β, f_c and Z_{TM}.
Network analysis				
8	19-20 3(c)	5	Signal flow graphs	Draw a node for each wave; add a branch for each path; apply the self-loop rule to find b/a, then Z_{in}.
RF components and filters				
9	19-20 5(b)	6	Filters	Treat the RC filter as a voltage divider; write V_o/V_i; find its size and phase; read off the cut-off.
10	18-19 5(a)	6	Two-port parameters	Write the matrix of the series C and of the shunt L; multiply them.
Microwave systems				
11	18-19 2(c)	5	Radio link budget	Spread the power over a sphere; collect it with the aperture; divide by the noise power $kT_{sys}B$.
12	18-19 6(c)	6	Radio link budget	Density at the receiver with $D = 4\pi A/\lambda^2$; multiply by the receiving aperture.

No.	Paper	Marks	Derivation	Outline
Antennas				
13	20-21 4(a)	8	Hertzian dipole	Take the curl of A in spherical coordinates; divide by μ to get H.
14	20-21 4(b)	12	Hertzian dipole	Use the curl of H in free space; divide by j$\omega\varepsilon$; collect the r and θ components of E.
15	20-21 4(c)	5	Hertzian dipole	Keep the 1/r far-field terms; form the Poynting vector E $\times$ H*; average it over time.
16	20-21 6(b)	13	Radiation from accelerating charges	Start from the retarded potentials; take E and B from them; keep the 1/R term, which needs an accelerating charge.
17	19-20 5(c)	7	Linear arrays	Add the fields of N elements; sum the geometric series; set $\psi = 0$ for the main beam and $\psi = 2\pi$k/N for nulls.
18	18-19 1(d)	8	Magnetic dipole	Find **A** of the loop by Stokes; curl it for **H**; curl **H** for **E**.
19	18-19 2(d)	9	Linear arrays	Add the fields of N elements; sum the geometric series; set $\psi = 0$ for the main beam and $\psi = \pm 2k\pi/N$ for nulls.
20	18-19 6(b)	6	Paraboloid reflectors	Write r and ρ in terms of θ; balance the power in a cone and a ring; take the ratio to the centre.
21	18-19 6(d)	7	Horn antennas	Put $A_e = 0.6A_p$ in $D = 4\pi A_e/\lambda^2$.

6. Repeat tracker:

Questions that came back, exactly or with new numbers.

Paper	Asked before as	Topic	Subtopic
2020-21 2(a)	2019-20 2(a)	Smith chart	Constant-r circles
2020-21 2(b)	2019-20 4(d)	Impedance matching	Single-stub matching
2020-21 2(c)	2019-20 4(b)	Impedance matching	Quarter-wave transformer
2020-21 6(d)	2019-20 1(a)	Antennas	Effective height and area
2018-19 1(a)	2019-20 1(b)	Antennas	Antenna aperture
2018-19 1(b)	2019-20 1(d)	Antennas	Pattern calculations
2018-19 1(c)	2019-20 5(d)	Antennas	Near and far field
2018-19 2(a)	2019-20 1(c)	Antennas	Remote sensing
2018-19 2(b)	2019-20 4(c)	Microwave systems	Radar
2018-19 2(d)	2019-20 5(c)	Antennas	Linear arrays
2018-19 3(d)	2019-20 4(d)	Impedance matching	Single-stub matching
2018-19 4(b)	2020-21 6(a)	Waveguides and microstrip	Microstrip
2018-19 4(d)	2019-20 4(b)	Impedance matching	Quarter-wave transformer

7. Question style:

How each topic was asked, as a number of questions.

Topic	Theory	Derivation	Numerical	Diagram	Design
EM waves	·	·	1	·	·
Transmission lines	3	4	6	·	·
Smith chart	2	2	·	·	·
Waveguides and microstrip	3	1	1	·	·
Network analysis	1	1	·	·	·
Impedance matching	3	·	·	·	4
Resonators	1	·	1	·	·
RF components and filters	1	2	·	1	·
Amplifier design	3	·	·	·	1

Free shared copy · Nur Mohammad

Topic	Theory	Derivation	Numerical	Diagram	Design
Microwave tubes and devices	4	·	·	·	·
Microwave systems	3	2	3	·	·
Antennas	12	9	2	·	·

8. Study order:

Topics by marks offered, with the share of all marks covered so far. Papers offer more marks than you answer, so this is a guide to where marks have been, not a score.

No.	Topic	Marks	Covered so far
1	Antennas	145	32 %
2	Transmission lines	86	51 %
3	Impedance matching	43	61 %
4	Microwave systems	42	70 %
5	Smith chart	28	76 %
6	Waveguides and microstrip	27	82 %
7	Amplifier design	26	88 %
8	RF components and filters	24	93 %
9	Network analysis	11	96 %
10	Microwave tubes and devices	9	98 %
11	EM waves	6	99 %
12	Resonators	5	100 %

Free shared copy · Nur Mohammad